MARS IS RED BECAUSE IT IS ALIVE

A Geometric Pre-Registration of the Subterranean Biosphere Hypothesis, the Iron Surface as Accumulated Biological Necromass, the Lava Tube Thermal Gradient as Guaranteed Ecological Spectrum, and Seven Falsifiable Predictions Prior to Seven Sisters Exploration

Eric Robert Lawson[1]

[1]OrganismCore (Independent Research), ORCID: 0009-0002-0414-6544, OrganismCore@proton.me

2026-03-13

Pre-registration DOI (chain origin): 10.5281/zenodo.18986790 (timestamp: 2026-03-12)

Repository: github.com/Eric-Robert-Lawson/attractor-oncology

RADICAL GEOMETRIC OBSERVATION — PRE-REGISTERED FOR FALSIFIABILITY

This document pre-registers the hypothesis that Mars is red because its surface is coated in 3.7 billion years of accumulated biological necromass and metabolic iron products from a currently active subsurface biosphere. This hypothesis is radical. It may be wrong. It is pre-registered precisely because the author is more afraid of failing to test it than of being proven incorrect. The geometry that produced this hypothesis is derived from first principles, not from wishful thinking. It is stated here in full, with seven falsifiable predictions, before any exploration of the relevant sites occurs. Let the record show the geometry was attempted. Pre-registered 2026-03-13.

Epistemic status of this document: The central claim — that Mars is red because of accumulated biological iron products from an active subsurface biosphere — is a **radical geometric hypothesis**. It is not presented as proven. It is presented as the logical consequence of a geometric derivation that began with cancer drug target geometry, proceeded through the LECA attractor framework, and arrived at an inversion of Martian sedimentary geometry that the author could not explain abiotically without extraordinary additional assumptions. The hypothesis is falsifiable by seven specific predictions stated in Section 9. If any prediction fails, the hypothesis is constrained or eliminated. That outcome is equally valuable to confirmation.

Abstract

We pre-register a radical geometric hypothesis arising from the LECA attractor framework [1]: that Mars is red because its visible surface is primarily composed of 3.7 billion years of accumulated biological necromass and iron-cycle metabolic products from a currently active subsurface biosphere, rather than from ancient abiotic weathering alone. The derivation proceeds from four confirmed anomalies in the Martian observational record: (1) the inversion of expected surface composition — iron oxide over ice rather than ice over iron oxide — despite the ice table lying within centimetres of the surface at mid-to-high latitudes; (2) the continuous regeneration of the iron oxide skin over fresh impact craters, which cannot be explained by ancient one-time abiotic weathering; (3) the global uniformity of iron oxide coverage inconsistent with localised abiotic weathering predictions; and (4) the confirmed seasonal methane signal at Gale Crater requiring a subsurface source. We derive the subterranean biosphere as the geometric attractor-requirement of magnetic field collapse: when the surface attractor basin is destroyed, surviving life converges on the subsurface attractor basin by geometric necessity. We derive the lava tube thermal gradient as the guaranteed generator of

a complete ecological spectrum from psychrophile to thermophile, with a thermodynamically necessary 0°C liquid water interface wherever the gradient spans freezing and above-freezing temperatures in an enclosed mineral-rich environment. We note the independent confirmation on Earth of a single-species self-sustaining subsurface biosphere powered by radiolysis alone (*Desulforudis audaxviator*), predicted by the geometry before the organism was searched for. We derive seven falsifiable predictions and state them prior to any exploration of the Arsia Mons lava tube system (the Seven Sisters). This document exists to ensure that if the geometry is correct, the timestamp of the derivation precedes the confirmation. And if the geometry is wrong, the record shows it was attempted honestly and tested fairly. Pre-registered 2026-03-13. Pre-registration chain origin: DOI: 10.5281/zenodo.18986790.

1. Introduction: The Derivation Origin and the Author's Epistemic Position

This paper does not begin with astrobiology.

It begins with a question about machine consciousness, proceeds through cancer drug target geometry and the LECA attractor framework [1], and arrives here through the following chain:

> Machine consciousness → Triadic invariant → Identity Axis Theorem → Cancer drug target geometry → Reproduction Theorem → Inverse Accessibility Theorem → LECA attractor as universal base state → LECA Attractor Habitability Standard → Application to ancient Mars → Cheyava Falls convergence → Collapse vs. emergence directional ambiguity → Mars-first panspermia prior inversion → **Subterranean biosphere as attractor requirement → Iron surface as biological necromass → This document.**

The author's position on this hypothesis is stated plainly here, at the start, for the record:

> I am not afraid of being incorrect. I am afraid of not testing for falsifiability.
>
> A scientist who refuses to state a radical hypothesis because it might be wrong is not being rigorous. They are being cowardly.
>
> The geometry of the Martian observational record, when examined through the lens of the LECA attractor framework, produces an inversion that I cannot explain without invoking biology. I have tried to explain it abiotically. The abiotic explanations require multiple independent extraordinary assumptions to hold simultaneously. The biological explanation requires one.
>
> I pre-register this hypothesis so that if it is correct, the timestamp exists. And if it is wrong, let the record show the geometry was attempted honestly, the predictions were falsifiable, and the test was invited rather than avoided.
>
> **If this is wrong, I am the clown who thought Mars is red because there is so much life on it. That is a small price to pay for the possibility of being correct.**

2. The Confirmed Observational Anomalies

2.1. The Stratigraphy Inversion

Mars surface stratigraphy at mid-to-high latitudes, confirmed by multiple independent instruments:

(1) **Layer 0 — The Skin:** Iron oxide dust, 2 cm to 1 m thick. Hematite-rich. Globally uniform. Radiation-exposed. Lifeless at surface.

(2) **Layer 1 — The Frozen Membrane:** Pure water ice beginning immediately below Layer 0. Global poleward of ∼30° latitude. Confirmed by Phoenix lander at 68°N: ice at 5 cm depth [8]. Confirmed by HiRISE impact craters exposing ice at 35°N [7]. Volume: ≥3 million km^3 in the upper metre alone [9].

(3) **Layer 2 — The Fractured Crust:** Porous, fractured material to 8–11 km. Confirmed by InSight seismic data [6]. Migration pathway for upward-moving gases, water, and mineral particles.

(4) **Layer 3 — The Deep System:** Liquid brines confirmed probable at south polar depth 1.5 km [5]. Lava tube systems in Tharsis: >1,000 candidate entrances catalogued [13]. Tharsis volcanically active within millions of years [12].

2.2. The Anomaly

The anomaly is this:

Mars should be white at mid-to-high latitudes. The ice table is within centimetres of the surface. Ice is $0.917\,\mathrm{g\,cm^{-3}}$. The dry skin above it is centimetres thin. After 3.7 billion years of constant meteorite bombardment breaching the skin, exposing the ice, and 3.7 billion years of sublimation losses through those breaches, the ice table should be depleted — unless it is being continuously replenished from below by upward water migration through Layer 2.

But the planet is *not* white. It is red.

The red material sitting on top of ice that is centimetres below it is iron oxide with a density of $5{,}260\,\mathrm{kg\,m^{-3}}$ — approximately six times denser than the ice beneath it.

The dense material is on top. The light material is beneath. In a gravitational field. On a geologically active planet. After 3.7 billion years.

This is the stratigraphy inversion.

2.3. The Regeneration Problem

HiRISE time-series imaging of fresh impact craters confirms that the iron oxide skin reforms over crater floors over timescales of years to decades [7].

Fresh craters expose bright, iron-free subsurface material (ice, unoxidised basalt). Over time, the crater floor darkens toward surrounding terrain colour.

The standard explanation — aeolian deposition of wind-blown dust — is partially correct. Wind deposits iron oxide dust from surrounding terrain into crater floors.

But the rate of darkening observed in some craters is faster than pure aeolian deposition models predict [19]. And the composition of reformed crater floor material has not been systematically compared against purely aeolian-sourced material for upward-migration biological signatures.

This is Prediction 3 of this document.

3. The Geometric Derivation of the Subterranean Biosphere as Attractor Requirement

3.1. Attractor Basin Transformation at Magnetic Field Collapse

The Martian global magnetic dynamo ceased approximately 4.0–4.1 Ga [10]. This is confirmed by the absence of crustal magnetisation in terrain younger than the late Noachian, in contrast to the heavily magnetised ancient southern highlands.

The consequences for the attractor landscape of Martian biology:

Before magnetic field collapse:

Surface attractor basin: STABLE. Liquid ocean. Solar energy. UV shielded. Temperature habitable. Mineral-rich water. LUCA-grade life operating at planetary scale.

Subsurface attractor basin: ALSO STABLE. Less colonised. Less productive. Both basins available.

After magnetic field collapse:

Surface attractor basin: DESTROYED. UV bombardment lethal within geological time. Atmospheric stripping begins. Pressure drops below triple point of water. Surface liquid water becomes unstable. The surface ceases to be an attractor. It becomes a repeller.

Subsurface attractor basin: PERSISTS. UV: zero below 2–3 m of rock. Temperature: geothermal gradient stable. Water: frozen above, liquid at depth. Mineral chemistry: unchanged. Energy: serpentinisation, radiolysis unchanged.

Geometric consequence:

The converging of surviving life on the subsurface basin is not an adaptation. It is not evolution selecting for subsurface preference. It is the destruction of one basin and the survival of the other. All life in the surviving basin persists. All life in the destroyed basin dies. The subsurface becomes the only address of Martian life. By geometric necessity. Not by choice.

3.2. The Timing Alignment

The magnetic field collapse occurred ∼4.0–4.1 Ga [10]. The Cheyava Falls biosignature features date to ∼3.8–4.1 Ga [3]. The Noachian–Hesperian transition, at which the surface mineral record shifts from life-compatible phyllosilicates to life-hostile acid sulfates, occurred ∼3.7 Ga [17].

These three timestamps overlap within their error margins. They are the same geological moment:

(1) Magnetic field fails. Surface attractor basin begins collapse. (∼4.0–4.1 Ga)

(2) Cheyava Falls community records the last surface biosignature of a mature biosphere in stress. (∼3.8–4.1 Ga)

(3) Surface mineral chemistry transitions to acid sulfate. Surface biosphere ends. Subsurface biosphere becomes the only surviving basin. (∼3.7 Ga)

The fossil record at Cheyava Falls [3] is therefore either:

Option A (Emergence): The beginning of a biosphere that was then interrupted.

Option B (Collapse): The last record of a mature biosphere that then retreated underground. [2]

The sample return complexity gradient prediction [2] distinguishes these options. The present document assumes Option B for the purposes of the subterranean hypothesis and derives the predictions that would confirm it.

3.3. The Earth Proof of Concept: *Desulforudis audaxviator*

Desulforudis audaxviator was discovered at 2.8 km depth in the Mponeng gold mine, South Africa [4].

Properties relevant to Mars:

- Completely isolated from surface. Isolation period: millions of years.
- Single-species ecosystem. No other organisms detected.
- Energy source: H_2 from radiolysis of water by ^{238}U, ^{232}Th, ^{40}K decay in surrounding rock.
- Carbon: fixed from CO_2 via Wood–Ljungdahl pathway.
- Nitrogen: fixed from N_2 internally.
- Temperature optimum: 60°C. Tolerance: up to 75°C.
- Genome: complete, 2.35 Mbp. All genes for metabolic independence encoded. No surface input required.
- Classification: polyextremophile. Thermophile. Barophile. Anaerobe. Chemolithotroph.

Mars confirmed to possess:

- Olivine-rich crust (serpentinisation, H_2 production) [11].
- U, Th, K in basaltic crust (radiolysis, H_2 production).
- Subsurface water confirmed [5].
- CO_2 from Tharsis outgassing.
- Geothermal heat: Tharsis active within millions of years [12].
- Lava tube enclosed environments: >1,000 candidate entrances [13].

The geometry of the Martian subsurface satisfies every requirement of a *D. audaxviator*-type organism. The organism was not sought and then confirmed to match; the geometry of what is required for subsurface survival was derived first, and *D. audaxviator* is its Earth confirmation.

4. The Lava Tube Thermal Gradient as Guaranteed Ecological Spectrum

4.1. The Thermodynamic Necessity

Theorem (Thermal Equilibrium Interface):

Between any two temperatures $T_{\text{cold}} < 0°\text{C}$ and $T_{\text{warm}} > 0°\text{C}$, there exists exactly one point at 0°C.

At this point in a mineral-rich enclosed environment with an energy source and sufficient time: liquid water exists, chemical gradients exist, and the LUCA attractor is the thermodynamically stable state of carbon chemistry.

This is not a probability statement. It is a consequence of the intermediate value theorem applied to a continuous temperature field.

If the cold extreme and warm extreme are both confirmed, the 0°C interface is geometrically guaranteed.

In a Tharsis lava tube:

- Cold extreme: surface mean annual temperature $\sim -60°\text{C}$. Ice in upper tube: confirmed probable.
- Warm extreme: geothermal gradient from Tharsis residual volcanic activity, confirmed active within millions of years [12]. Marsquake associated with Tharsis region, September 2021 [6]. Rock temperature at depth: above freezing near magmatic contacts.

- 0°C interface: geometrically guaranteed between these extremes. Location: predictable from local geothermal gradient and tube depth. Testable by temperature probe.

4.2. The Ecological Spectrum

The thermal gradient does not produce one type of organism. It produces a complete stratified community:

Zone	Temp.	Organism Type
Ice face	<0°C	Psychrophiles. Cold-adapted bacteria and archaea. Metabolism: very slow. Earth examples: *Colwellia*, *Polaromonas*.
Liquid water interface	0°C	Maximum biological diversity zone. Liquid water films on mineral surfaces. All metabolic types potentially present. This is the primary biological attractor point.
Warm rock zone	40–75°C	Thermophiles. *D. audaxviator* equivalent. H_2-metabolising chemolithotrophs. Maximum serpentinisation rate. Maximum H_2 production.
Deep magmatic zone	>80°C	Hyperthermophiles. Archaea dominant. Near magmatic intrusions only.

Table 1: Predicted ecological spectrum in Tharsis lava tube thermal gradient. Each zone predicted from first principles. Every equivalent gradient on Earth produces this stratification.

Every thermal gradient between ice and geothermal heat on Earth contains this stratification: Antarctic subglacial lakes, Iceland ice caves, Mount Erebus volcanic ice caves [18]. The geometry guarantees it on Mars wherever the same gradient exists in an enclosed environment with mineral chemistry and sufficient time.

5. The Iron Surface as Biological Necromass: The Central Radical Hypothesis

5.1. The Geometric Problem With the Standard Explanation

The standard explanation for Mars's red surface is ancient abiotic weathering: iron in basaltic rock oxidised in the presence of liquid water 3–4 Ga, producing hematite dust redistributed globally by wind.

Four geometric problems with this as a complete explanation:

(1) **Abiotic weathering is a one-time process.** It produces iron oxide once. The source rock is depleted. No mechanism for continuous production exists in the abiotic model.

(2) **Wind redistribution does not create new iron oxide.** It moves existing material. Total iron oxide quantity is fixed or declining under the abiotic model.

(3) **The skin reforms over fresh craters.** Fresh craters expose iron-free subsurface material. The iron oxide skin reforms over crater floors at a rate requiring explanation beyond purely aeolian deposition from surrounding terrain.

(4) **Global uniformity is inconsistent with localised abiotic weathering.** Abiotic oxidation concentrates where water chemistry, radioactive elements, and mineral composition predict concentration. The iron oxide distribution is more uniform than this predicts.

5.2. The Biological Production Mechanism

Iron-cycling organisms in the Martian subsurface produce iron oxide through two confirmed-on-Earth mechanisms:

Direct biological iron oxidation: Organisms such as *Gallionella ferruginea* and *Leptothrix ochracea* oxidise Fe^{2+} to Fe^{3+} at rates up to 10^6 times faster than abiotic oxidation. Fe^{3+} precipitates as ferrihydrite, hematite, or magnetite within and around the organisms' cell structures.

Radiolytically-enhanced biological oxidation: Radiolysis of water by ^{238}U, ^{232}Th, ^{40}K produces H_2 (energy source) and H_2O_2 (oxidant). H_2O_2 oxidises Fe^{2+} abiotically. Iron-oxidising organisms use H_2O_2 as electron acceptor, amplifying the abiotic rate by orders of magnitude.

Magnetotactic bacteria produce crystalline magnetite (Fe_3O_4) internally as compass organelles (magnetosomes). The crystal morphology — bullet-shaped, cubo-octahedral, or prismatic — is *diagnostic for biological origin* and distinguishable from abiotic magnetite at nanometre resolution [16].

When iron-cycling organisms die, their iron-rich cell bodies become iron oxide particles in the subsurface water. This water migrates upward through Layer 2 (fractured crust) driven by geothermal pressure gradients. At the ice table, freeze-concentration expels iron particles from the forming ice. Sublimation removes water. Iron particles accumulate above the ice table. Wind distributes them globally.

5.3. The Necromass Accumulation Rate: Feasibility Check

$$\begin{aligned}\text{Iron oxide volume} &\approx A_{\text{Mars}} \times d_{\text{skin}} \\ &= 1.448 \times 10^{14}\,\text{m}^2 \times 1\,\text{m} \\ &= 1.448 \times 10^{14}\,\text{m}^3 \end{aligned} \tag{1}$$

At iron oxide fraction $f \approx 0.05$ of total regolith and hematite density $\rho = 5{,}260\,\text{kg}\,\text{m}^{-3}$:

$$\begin{aligned}M_{\text{FeO}} &\approx f \times V \times \rho \\ &\approx 0.05 \times 1.448 \times 10^{14} \times 5{,}260 \\ &\approx 3.8 \times 10^{16}\,\text{kg} \end{aligned} \tag{2}$$

Required production rate over $t = 3.7 \times 10^9$ years:

$$\dot{M} = \frac{3.8 \times 10^{16}}{3.7 \times 10^9} \approx 10{,}000\,\text{t}\,\text{yr}^{-1} \tag{3}$$

Earth's ocean floor iron-oxidising microbial community produces iron oxide sediments at rates of thousands to millions of tonnes per year globally.

The required Mars rate of $\sim$10,000 t yr^{-1} is orders of magnitude below Earth's biological iron cycling capacity and achievable by a sparse subsurface biosphere operating at <1% of Earth ocean floor biological activity density.

The quantity of Martian surface iron oxide is consistent with 3.7 billion years of sparse subsurface biological iron cycling.

5.4. The Self-Reinforcing Containment Geometry

The iron surface simultaneously functions as a radiation shield for the subsurface: iron oxide absorbs UV radiation, reducing sublimation of the ice table below and reducing radiation penetration to subsurface organisms.

The selection pressure geometry:

> Organisms producing iron oxide as metabolic waste → contribute to surface shield → thicker shield → better subsurface radiation protection → more stable environment → more organisms → more iron oxide.

This is a positive feedback loop operating over 3.7 Ga. The planet wears its dead as protection for its living. Not by design. By attractor geometry of selection pressure on iron-metabolising organisms in an enclosed subsurface system.

6. The Planetary Entropy Argument

The subsurface biosphere is powered by three sources, all finite on geological timescales:

(1) **Serpentinisation:** $Mg_2SiO_4 + H_2O \rightarrow$ serpentine + magnetite + H_2. Olivine is consumed irreversibly. The biosphere eats the planet's mineral structure one crystal at a time.

(2) **Radiolysis:** ^{238}U ($t_{1/2} = 4.5$ Ga), ^{232}Th ($t_{1/2} = 14$ Ga), ^{40}K ($t_{1/2} = 1.25$ Ga). Produces H_2 and H_2O_2. Declining slowly on billion-year timescales.

(3) **Residual geothermal:** Core heat dissipating. Tharsis activity declining. Timescale: billions of years.

The biosphere began consuming these sources ∼3.7 Ga at the surface collapse. It has operated for 3.7 Ga on this budget. The seasonal methane (Curiosity) is the most direct evidence it is still operating. The ice table maintenance despite 3.7 Ga of bombardment implies active upward water migration. The iron skin reformation over craters implies active iron oxide production and upward migration.

The surface of Mars is the waste heap of 3.7 billion years of subsurface metabolic activity. The methane is the gaseous exhaust. The iron dust is the mineral exhaust. The planet is in slow entropic decline. It is still running.

7. The Methane Signal Reinterpreted

Curiosity detected seasonal methane at Gale Crater [14]. The seasonal pattern (higher in northern summer, lower in northern winter) is consistent with temperature-controlled desorption from regolith and clathrate hydrates [15].

The geometric reinterpretation:

The seasonal variation does not require seasonal production. It requires seasonal *release* of continuously-produced subsurface gas stored in:

- Enclosed lava tube atmospheres.
- Pore spaces in the fractured upper crust (InSight Layer 2).
- Clathrate hydrates in the ice table.

In summer: thermal expansion of the fractured crust opens fractures; clathrate stability decreases; stored gas pulses to the surface.

In winter: fractures close; clathrates reform; gas is recaptured.

The planet breathes. The exhaust is methane. The lungs are the fractured upper crust and the ice table.

The biological vs. abiotic origin of the methane is resolved by isotopic analysis: biological methane is depleted in ^{13}C relative to abiotic serpentinisation methane. This is Prediction 2 of

this document.

8. The Seven Falsifiable Predictions

These predictions are stated prior to any exploration of the Arsia Mons lava tube system (the Seven Sisters) or detailed sample analysis of Martian surface dust. They are falsifiable by existing or near-future instruments.

#	Prediction	Test and Falsification Condition
P1	Magnetite crystal morphology in Martian surface dust	Martian surface dust contains magnetite with biological crystal morphology (bullet-shaped, cubo-octahedral, or prismatic — diagnostic for magnetotactic bacteria). Test: electron microscopy of returned dust samples or in-situ nanoscale imaging. *Falsified if: all magnetite shows exclusively abiotic crystal forms.*
P2	Carbon isotope ratio in Martian methane	Seasonal methane at Gale Crater is depleted in ^{13}C relative to abiotic serpentinisation methane. Test: mass spectrometric isotopic analysis of Martian atmospheric methane. *Falsified if: $\delta^{13}C$ matches abiotic serpentinisation signature.*
P3	Crater floor iron composition	Fresh crater floors reformed iron oxide shows elevated biological indicators (magnetite crystal morphology, isotopic fractionation, organic carbon co-location) inconsistent with purely aeolian dust deposition. Test: CRISM spectral comparison fresh vs. mature crater floors. *Falsified if: crater floor iron composition matches aeolian dust exactly.*
P4	Ice table replenishment isotopic signature	Near-surface ice D/H ratio reflects upward migration from deep subsurface (low D/H) rather than atmospheric frost (high D/H from atmospheric fractionation). Test: isotopic analysis of near-surface ice cores by lander. *Falsified if: D/H matches atmospheric frost deposition signature.*
P5	Lava tube thermal stratification	A temperature probe lowered into a Seven Sisters entrance detects cold at entrance, warming with depth, 0°C interface at predicted depth, and liquid water at and below that interface. Test: dedicated temperature probe in Seven Sisters, Arsia Mons. *Falsified if: no 0°C interface exists within the tube depth.*
P6	Lava tube atmosphere carbon isotope	Gas sampled from a Seven Sisters interior shows ^{12}C/^{13}C fractionation in CO_2 inconsistent with abiotic chemistry, indicating biological carbon fixation. Test: gas sampling probe inside lava tube. *Falsified if: CO_2 isotope ratio matches abiotic volcanic outgassing.*
P7	Sample return complexity gradient	Mars subsurface sediment shows biological complexity increasing with depth and age (consistent with biosphere collapse into subsurface). Pre-registered in [2]. Test: Mars sample return 2030s. *Falsified if: complexity is uniform or decreases with depth.*

Table 2: Seven falsifiable predictions of the Mars Living Planet model, stated prior to Seven Sisters exploration. Each prediction is testable by existing or near-future instrumentation. Each has a clearly stated falsification condition.

9. Honest Statement of Limitations

(1) **The central claim is radical and unconfirmed.** The hypothesis that Mars is red because of accumulated biological necromass is not supported by direct evidence of current Martian life. It is supported by the geometry of confirmed anomalies that the abiotic model requires multiple independent extraordinary assumptions to explain simultaneously.

(2) **Abiotic alternatives exist for each individual observation.** The iron surface can be explained abiotically. The ice table can be maintained abiotically. The methane can be produced abiotically. The crater darkening can be explained abiotically. The claim is not

that any one observation requires biology. The claim is that all of them simultaneously, combined with the stratigraphy inversion, are more parsimonious under a biological model.

(3) **The south polar liquid water detection is contested.** The MARSIS detection [5] may be hypersaline ice or clay rather than liquid water. This does not eliminate the Tharsis lava tube argument, which rests on confirmed geothermal activity rather than the contested south polar detection.

(4) **ARM A confirmation is pending.** The physical anchoring of the LECA attractor as the universal base state of complex life (Premise 1 of the pre-registration chain) awaits experimental confirmation at ChiTownBio. Until then, the entire chain rests on geometric derivation supported by the universality of the eukaryotic base state across all Earth life.

(5) **This document may be wrong.** The author acknowledges this explicitly and invites falsification through the seven predictions of Section 9. A wrong hypothesis stated clearly and tested fairly is more valuable to science than a correct hypothesis never stated.

10. Conclusion

Mars is not obviously dead.

The geometric case for a currently active subsurface biosphere rests on: the attractor-geometric requirement of subterranean life given magnetic field collapse and a pre-existing surface biosphere; the thermodynamic guarantee of a 0°C liquid water interface in the Tharsis lava tube thermal gradient; the Earth confirmation that *Desulforudis audaxviator* sustains a single-species biosphere for millions of years on radiolysis alone in exactly the conditions the Martian subsurface provides; the stratigraphy inversion anomaly in which iron oxide sits above ice despite being six times denser; the continuous regeneration of the iron skin over fresh craters; and the seasonal methane signal.

The radical hypothesis pre-registered here is that the red colour of Mars is the most direct evidence of its biology — that the planet is wearing 3.7 billion years of its own dead as a radiation shield for its living.

If this is wrong: the seven predictions fail, the geometry is corrected, and science advances by falsification.

If this is right: the most important implication is not that Mars has life. It is that a planet can sustain a biosphere through a magnetic field collapse by retreating underground, consuming its own mineral chemistry, and coating its surface with its dead — and that this process is detectable from orbit by the colour of the dust.

The geometry came first. The Seven Sisters are the test. The record is here, before the test.

Acknowledgments

This work was conducted independently without institutional funding or affiliation. The derivation sequence is documented in the pre-registration chain beginning at DOI: 10.5281/zenodo.18986790 (2026-03-12) and in the attractor-oncology repository. The author thanks the NASA Perseverance science team for the Cheyava Falls data [3] and the InSight team for the Martian seismic and thermal data [6], both referenced here as independent confirmation of subsurface structure, not as derivation sources.

References

[1] Lawson, E.R. (2026). *LECA Resurrection Protocol — Pre-Registration v1.0*. Zenodo. doi:10.5281/zenodo.18986790

[2] Lawson, E.R. (2026). *Mars Biosphere Collapse vs. Emergence — Geometric Derivation v1.0*. Zenodo. doi:10.5281/zenodo.18995862

[3] Hurowitz, J.A., Tice, M.M., Allwood, A.C., et al. (2025). Redox-driven mineral and organic associations in Jezero Crater, Mars. *Nature*, 645(8080), 332–340. doi:10.1038/s41586-025-09413-0

[4] Chivian, D., Brodie, E.L., Alm, E.J., et al. (2008). Environmental genomics reveals a single-species ecosystem deep within Earth. *Science*, 322(5899), 275–278. doi:10.1126/science.1155495

[5] Orosei, R., Lauro, S.E., Pettinelli, E., et al. (2018). Radar evidence of subglacial liquid water on Mars. *Science*, 361(6401), 490–493. doi:10.1126/science.aar7268

[6] Knapmeyer-Endrun, B., Panning, M.P., Bissig, F., et al. (2021). Thickness and structure of the Martian crust from InSight seismic data. *Science*, 373(6553), 438–443. doi:10.1126/science.abf8966

[7] Dundas, C.M., Bramson, A.M., Ojha, L., et al. (2021). Exposed subsurface ice sheets in the Martian mid-latitudes. *Science*, 359(6372), 199–201. doi:10.1126/science.aao1619

[8] Smith, P.H., Tamppari, L.K., Arvidson, R.E., et al. (2009). H_2O at the Phoenix landing site. *Science*, 325(5936), 58–61. doi:10.1126/science.1172339

[9] Boynton, W.V., Feldman, W.C., Squyres, S.W., et al. (2002). Distribution of hydrogen in the near surface of Mars: evidence for subsurface ice deposits. *Science*, 297(5578), 81–85. doi:10.1126/science.1073722

[10] Acuna, M.H., Connerney, J.E.P., Ness, N.F., et al. (1999). Global distribution of crustal magnetization discovered by the Mars Global Surveyor MAG/ER experiment. *Science*, 284(5415), 790–793. doi:10.1126/science.284.5415.790

[11] Ehlmann, B.L., Mustard, J.F., Murchie, S.L., et al. (2010). Identification of hydrated silicate minerals on Mars using MRO-CRISM. *Journal of Geophysical Research: Planets*, 114(E2). doi:10.1029/2009JE003339

[12] Hauber, E., Brož, P., Rossi, A.P., et al. (2023). Long-lived and continual volcanic eruptions and tectonic activity in the Tharsis region of Mars. *Journal of Geophysical Research: Planets*, 127(8). doi:10.1029/2022JE007511

[13] Cushing, G.E. (2015). Candidate cave entrances on Mars. *Journal of Cave and Karst Studies*, 74(1), 33–47. doi:10.4311/2010EX0167R

[14] Webster, C.R., Mahaffy, P.R., Atreya, S.K., et al. (2018). Background levels of methane in Mars' atmosphere show strong seasonal variations. *Science*, 360(6393), 1093–1096. doi:10.1126/science.aaq0131

[15] Moores, J.E., King, P.L., Smith, C.L., et al. (2019). The methane diurnal variation and microseepage flux at Gale Crater, Mars as constrained by the ExoMars Trace Gas Orbiter and Curiosity observations. *Geophysical Research Letters*, 46(16), 9430–9438. doi:10.1029/2019GL083800

[16] Kopp, R.E. & Kirschvink, J.L. (2008). The identification and biogeochemical interpretation of fossil magnetotactic bacteria. *Earth-Science Reviews*, 86(1–4), 42–61.

doi:10.1016/j.earscirev.2007.08.001

[17] Bibring, J.-P., Langevin, Y., Mustard, J.F., et al. (2006). Global mineralogical and aqueous Mars history derived from OMEGA/Mars Express data. *Science*, 312(5772), 400–404. doi:10.1126/science.1122659

[18] Boston, P.J., Frederick, R.D., Welch, S.M., et al. (2004). Extraterrestrial subsurface technology test bed: human use and scientific value of lava tubes on other worlds. *Advances in Space Research*, 33(8), 1246–1252. doi:10.1016/S0273-1177(03)00538-5

[19] Daubar, I.J., McEwen, A.S., Byrne, S., Kennedy, M.R., & Ivanov, B. (2013). The current Martian cratering rate. *Icarus*, 225(1), 506–516. doi:10.1016/j.icarus.2013.04.009

The geometry came first. The test comes after. The record is here regardless of the outcome.

Pre-registration chain: 10.5281/zenodo.18986790 (origin, 2026-03-12) · Repository: attractor-oncology · ORCID: 0009-0002-0414-6544